Report R23 — Methylation and Drug Resistance in 187 Lung Cell Lines

A pre-registered test of the "reprogramming resistance through epigenetic modulation" thesis. The verdict was CONTRADICTED. Its own post-hoc control then reversed the sign, and neither reading is defensible.

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Summary

This replaces a set of figures that were entirely synthetic (see `results/SIDEQUEST-lung-epigenetic-figures/`) with measurements on real data. The question is the one that manuscript asserts: can epigenetic drugs be used against tumours that have become resistant to standard therapy?

Substrate. 198 lung cell lines from GSE68379, processed from raw 450k IDATs (485,512 probes), joined by COSMIC identifier to GDSC2 dose–response. **187 lines** carry both. Histology: 64 lung adenocarcinoma, 58 small cell, 21 mesothelioma, 14 squamous, 13 large cell, 11 NSCLC not otherwise specified, 4 carcinoid, 2 other.

The analysis plan — hypotheses, exclusions, gates, verdict thresholds and refusals — was written and SHA-256 hashed at `7e0322ba804aef2f...` *before any* IC_{50} *was joined to any methylation value*. The beta matrix was still being computed when the plan was hashed.

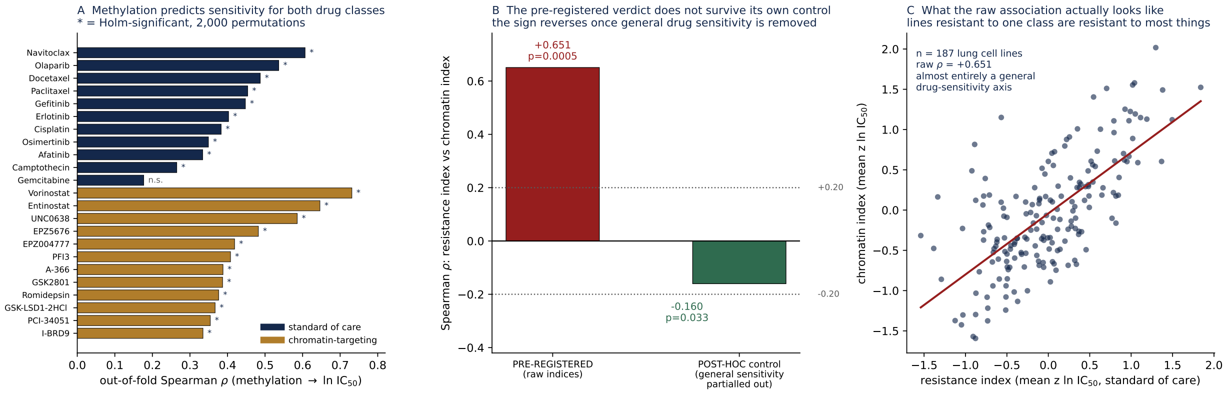
Three results

1. Methylation predicts drug sensitivity, and it is not subtle. Ten of eleven standard-of-care agents are predictable out-of-fold after Holm correction, and 14 of 32 chromatin-targeting drugs. Vorinostat reaches $\rho = 0.731$, Navitoclax 0.607, Olaparib 0.536, Docetaxel 0.487. Every EGFR inhibitor in the panel is predictable, Osimertinib included.

2. The pre-registered primary test says the thesis is wrong. Lines resistant to standard-of-care are *also* resistant to chromatin drugs: $\rho = +0.651$, permutation $p = 0.0005$, $n = 187$. The plan's rule maps $\rho \geq +0.20$ with $p < 0.05$ to **THESIS_CONTRADICTED**.

3. That verdict does not survive its own control. Both indices are mean z-scored $\ln \mathrm{IC}_{50}$, so both load on a general drug-sensitivity axis — some lines are simply resistant to nearly everything. Partialling out an index built from **227 GDSC2 drugs in neither set** takes the association from $+0.651$ to -0.160 ($p = 0.033$). The sign *reverses*, into the direction the thesis predicts, and lands below the pre-declared -0.20 threshold.

So the honest answer is that this design cannot settle the question. The pre-registered test measured a confound. The corrected estimate is weakly in the thesis's favour and too small to claim. Reporting either number alone would be misleading, and the plan did not anticipate the confound.



A Per-drug predictability, Holm-corrected over 2,000 permutations. **B** The primary test before and after the general-sensitivity control. **C** The raw association, which is what a general sensitivity axis looks like.

1 Standard-of-care agents

Drug	ρ	n	**Holm** p
Navitoclax	0.607	187	0.0055
Olaparib	0.536	187	0.0055
Docetaxel	0.487	187	0.0055
Paclitaxel	0.454	185	0.0055
Gefitinib	0.448	185	0.0055
Erlotinib	0.403	185	0.0055
Cisplatin	0.383	139	0.0055
Osimertinib	0.349	185	0.0055
Afatinib	0.334	187	0.0055
Camptothecin	0.265	187	0.0170
Gemcitabine	0.177	184	0.0555 *(n.s.)*

Top chromatin drugs: Vorinostat 0.731, Entinostat 0.646, UNC0638 0.586, EPZ5676 0.482, EPZ004777 0.419 — HDAC and histone-methyltransferase inhibitors respectively.

A caveat that applies to this whole table. These models may be predicting how sensitive a line is *in general* rather than its response to the specific drug. The same confound that destroyed the primary test is not controlled here, and the plan did not require it to be. The per-drug numbers should be read as "methylation carries information about this line's IC_{50}", not as "methylation predicts response to this agent specifically".

2 Two departures from the hashed plan

Both logged rather than silently applied.

The plan's permutation count made significance unreachable. It specified 200 permutations, giving a p-floor of $1/201 = 0.00498$. Holm across 11 standard-of-care drugs cannot then go below 0.0547, and across 32 chromatin drugs cannot go below 0.159. The first execution duly reported *zero* predictable drugs while Vorinostat sat at $\rho = 0.73$ — an arithmetic artefact of the plan, not a property of the data. Raised to 2,000 permutations, which can only make rejection easier and never harder, and gate G6 now fails loudly if the floor is ever too coarse for the family size again.

Sex-chromosome probes could not be excluded. Exclusion E5 required dropping them; the 450k manifest is not staged locally, so no probe was dropped on that criterion. 385,096 probes were retained after the missingness filter.

3 Controls

Gate	Check	Result
G0	plan digest matches the pre-registration	PASS
G1	187 lines with both methylation and GDSC2	PASS
G2	385,096 probes after filtering	PASS
PC1	small-cell vs non-small-cell separable, balanced accuracy 0.892	PASS
G3	all permutation nulls within 0.05 of zero (max 0.021)	PASS
G4	no drug retained below 100 lines (min 131)	PASS
G5	resistance and chromatin indices use disjoint drug sets	PASS
G6	permutation floor permits Holm significance	PASS

PC1 matters: it establishes the beta matrix carries real biology before any drug association is interpreted. Small-cell and non-small-cell lung lines separate at 0.892 balanced accuracy from methylation alone.

4 A data-integrity finding

Our mirror of `GSE68379_RAW.tar` is **corrupt and fails silently**. It is byte-complete against both GCS and GEO's declared size (9,020,579,840), ends with a valid tar terminator, and raises no error — but enumerates **1,302 of 2,056 members**, 649 of 1,028 cell lines, 127 of 198 lung. Two extraction modes gave the identical truncation.

It was caught only because the job asserted an expected array count and halted. The 396 lung IDATs were then fetched per sample from GEO instead: 396 of 396, zero failures. The data inventory describes this archive as 1,028 cell lines, which is true of GEO's copy and not of ours.

5 Limitations

Cell lines are not tumours. Methylation drifts in culture and $\ln \mathrm{IC}_{50}$ is a laboratory phenotype. Nothing here speaks to a patient, and the plan's refusals forbid saying otherwise.

The primary question is unresolved, not answered. A corrected pre-registration would build both indices on residuals after removing the general-sensitivity axis, and would declare that before looking.

187 lines against 485,512 probes. Every model is heavily regularised and only the out-of-fold estimate is reported.

Mixed histology. 58 of 187 are small cell and 21 mesothelioma, which differ from NSCLC in both methylation and drug response. No histology-stratified analysis was pre-declared.

Hash-before-label only. The plan author and the executor are the same person, so this is the weaker form of the protocol, not the structurally isolated form used in R16.

Reproducing this report

`evidence/PREREG_PLAN.yaml` (hashed `7e0322ba804aef2f...`), `evidence/PREREGISTRATION.json`, `evidence/run_analysis.py`, `evidence/process_idats.py` (GEO fetch and methylprep), `evidence/diagnose_t` (the archive diagnostic), `evidence/results.json`, `evidence/gates.json`, `evidence/resistance_index.npy`, `evidence/chromatin_index.npy`. Figure: `python3 make_figure.py`.